

MECHANICS

1. Kinematics (Motion in One Dimension)

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} a t^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x = x_0 + \frac{1}{2}(v_0 + v)t$$

Where: v_0 = initial velocity, v = final velocity

a = acceleration, t = time, x = position

2. Newton's Laws of Motion

First Law: An object at rest stays at rest, and an object in motion stays in motion at constant velocity, unless acted upon by a net external force.

Second Law: $F_{\text{net}} = m * a$ (vector equation)

Third Law: For every action, there is an equal and opposite reaction. $F_{AB} = -F_{BA}$

Common Forces:

Weight: $W = m * g$ ($g = 9.8 \text{ m/s}^2$ downward)

Normal Force: perpendicular to surface

Friction: $f_s \leq \mu_s * N$ (static)

$f_k = \mu_k * N$ (kinetic)

Tension: transmitted through strings/ropes

Spring: $F = -k * x$ (Hooke's Law)

3. Work, Energy, and Power

Work: $W = F * d * \cos(\theta)$

Kinetic Energy: $KE = \frac{1}{2} * m * v^2$

Potential Energy (gravity): $PE = m * g * h$

Potential Energy (spring): $PE = \frac{1}{2} * k * x^2$

Work-Energy Theorem: $W_{\text{net}} = \Delta(KE)$

Conservation of Energy: $KE_1 + PE_1 = KE_2 + PE_2$

Power: $P = W/t = F * v * \cos(\theta)$

4. Momentum and Collisions

Linear Momentum: $p = m * v$

Impulse: $J = F * \Delta(t) = \Delta(p)$

Impulse-Momentum Theorem: $J = m*v_2 - m*v_1$

Conservation of Momentum:

$$m_1*v_{1i} + m_2*v_{2i} = m_1*v_{1f} + m_2*v_{2f}$$

Elastic Collision: both KE and momentum conserved

Inelastic Collision: only momentum conserved

Perfectly Inelastic: objects stick together

5. Rotational Motion

Angular Velocity: $\omega = \Delta(\theta) / \Delta(t)$

Angular Acceleration: $\alpha = \Delta(\omega) / \Delta(t)$

Torque: $\tau = r * F * \sin(\theta) = I * \alpha$

Moment of Inertia: $I = \sum(m_i * r_i^2)$

Rotational KE: $KE_{\text{rot}} = (1/2) * I * \omega^2$

Angular Momentum: $L = I * \omega$

Conservation of L: $I_1*\omega_1 = I_2*\omega_2$

THERMODYNAMICS

6. Temperature and Heat

Temperature Scales:

$$F = (9/5)*C + 32$$

$$K = C + 273.15$$

Heat Transfer: $Q = m * c * \Delta(T)$

Where c = specific heat capacity

Phase Changes: $Q = m * L$

Where L = latent heat of fusion/vaporization

7. Laws of Thermodynamics

Zeroth Law: If A is in thermal equilibrium with B, and B with C, then A is in equilibrium with C.

First Law: $\Delta(U) = Q - W$

(change in internal energy = heat added - work done)

Second Law: Heat flows spontaneously from hot to cold.

Entropy of an isolated system always increases.

$$\Delta(S) \geq Q/T$$

Third Law: As $T \rightarrow 0$ K, $S \rightarrow 0$ for a perfect crystal.

8. Ideal Gas Law

$$PV = nRT$$

Where: P = pressure (Pa), V = volume (m^3)

n = moles, $R = 8.314 \text{ J/(mol}\cdot\text{K)}$, T = temp (K)

Kinetic Theory: $KE_{avg} = (3/2) * k_B * T$

Where $k_B = 1.38 * 10^{-23} \text{ J/K}$

WAVES AND SOUND

9. Wave Properties

Wave Speed: $v = f * \lambda$

Where f = frequency, λ = wavelength

Period: $T = 1/f$

Angular Frequency: $\omega = 2\pi f$

Wave Number: $k = 2\pi/\lambda$

Standing Waves (string): $f_n = n*v / (2*L)$

Standing Waves (pipe, open): $f_n = n*v / (2*L)$

Standing Waves (pipe, closed): $f_n = n*v / (4*L)$

Doppler Effect:

$$f_{observed} = f_{source} * (v \pm v_{observer}) / (v \mp v_{source})$$

ELECTRICITY AND MAGNETISM

10. Electric Charge and Force

Coulomb's Law: $F = k * |q_1*q_2| / r^2$

Where $k = 8.99 * 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$

Electric Field: $E = F/q = k*Q/r^2$

Electric Potential: $V = k*Q/r$

Electric Potential Energy: $U = k*q_1*q_2/r$

11. Circuits

Ohm's Law: $V = I \cdot R$

Power: $P = V \cdot I = I^2 \cdot R = V^2 / R$

Series: $R_{\text{total}} = R_1 + R_2 + R_3 + \dots$

Parallel: $1/R_{\text{total}} = 1/R_1 + 1/R_2 + 1/R_3 + \dots$

Capacitors in Parallel: $C_{\text{total}} = C_1 + C_2 + C_3 + \dots$

Capacitors in Series: $1/C_{\text{total}} = 1/C_1 + 1/C_2 + \dots$

Energy in Capacitor: $U = (1/2) \cdot C \cdot V^2 = (1/2) \cdot Q \cdot V$